Blueprint

The order generated by a column of the $SU(n)_k$ S-matrix is locally a numerical semigroup ring (Part II)

Algebraic number theory · orders and discriminants of fusion rings

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At a glance

What the paper contributes

Part I turned one column of the $SU(n)_k$ S-matrix into an order whose defect $\delta(T)$ is the number of gaps of a numerical semigroup S_T. This note asks how the columns fit together: the p-adic valuation of the discriminant of the fusion ring splits into ramification, defects and a gluing term whose only unknown piece is $\Delta(n,p) = \Sigma\, \delta$ over orbits. **That splitting is verified computationally, not proved**, as the paper says.

- **Small defects by vanishing moments** (Proposition 1, Corollary 3, Lemma 4): with trivial stabiliser, $\delta \geq 2 \Leftrightarrow M_2M_3 = 0$, and $\delta = 3$ happens in exactly four ways, one per numerical semigroup of genus three; $\delta = 2$ never occurs at $n = 4$. Observation 2 refines this and is measured.
- **The distribution law** (Theorem 5, Lemma 6, Theorem 8): for fixed $n \geq 5$ the proportion of columns with $\delta \geq 2$ is $2/p - 1/p^2$ up to an explicit error, with no factorial threshold; the mean defect per orbit is $1 + 2/p + 3/p^2$ for $n \geq 7$.
- **Uniformity in the central range** (Theorem 9, proof sketched): both expansions hold for $\varepsilon p \leq n \leq (1-\varepsilon)p$.
- **The two ends** (Propositions 10–11, §4): totally split trinomials $X^n + aX + b$ at the top (counted at $n = 5$ by an elliptic curve of conductor 50, measured); Gauss's half-systems in the middle, whose even moments vanish exactly and which give an infinite family of non-Gorenstein orders with semigroup $\langle 3,5,7\rangle$.

🚀 **Start here: Corollary 3**: with trivial stabiliser, $\delta \geq 2$ if and only if $M_2M_3 = 0$.

Main results:

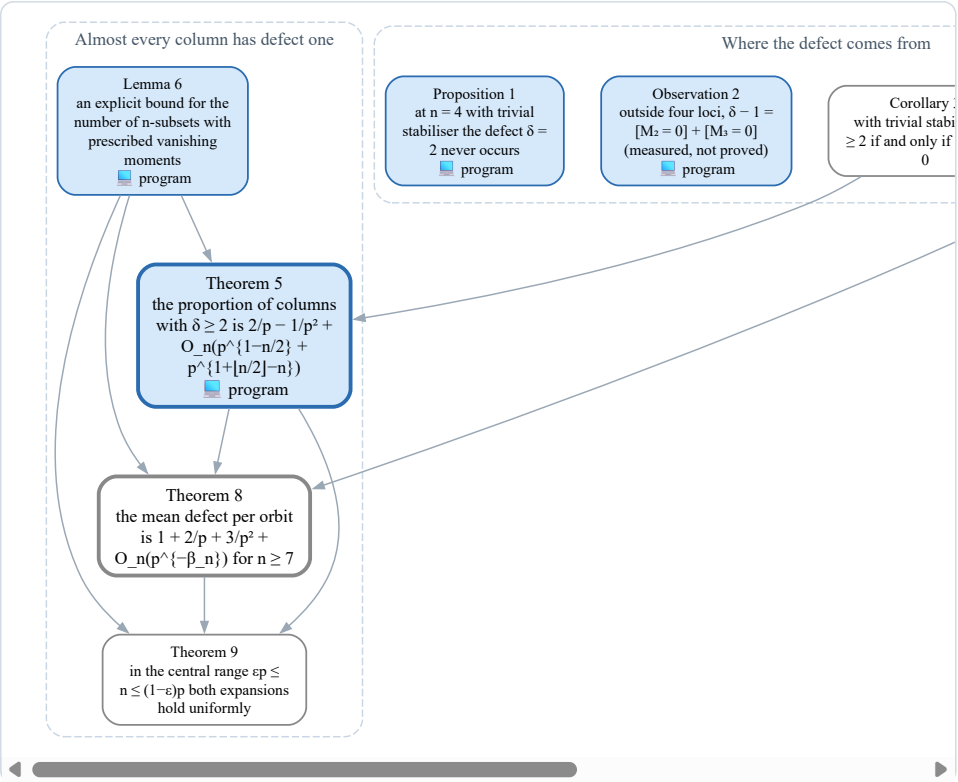
- **Lemma 4**: $\delta = 3$ in exactly four ways, one per numerical semigroup of genus three
- **Theorem 5**: the proportion of columns with $\delta \geq 2$ is $2/p - 1/p^2 + O_n(p^{1-n/2} + p^{1+\lfloor n/2\rfloor - n})$
- **Theorem 8**: the mean defect per orbit is $1 + 2/p + 3/p^2 + O_n(p^{-\beta_n})$ for $n \geq 7$
- **Proposition 11**: for every $p \equiv \pm 1 \pmod 8$, an explicit half-system gives a non-Gorenstein order with semigroup $\langle 3,5,7\rangle$

10 results | ✅ **0** formalized in Lean | ◐ **0** with a part in Lean

💻 **7** checked by a program | written proof only | (a result can carry several tags)

🗺️ Map of the results (click a box to go to it)

Dependencies in the written proof: an arrow A → B means that this proof of B uses A. Every result rests on its written proof; the tags record what was checked besides, and each statement's "Evidence & scope" block says how far. The colour of a box is its strongest tag, in the order above; the tags in the box give the rest. Boxes with a thick border are the main results; click a box to read the statement.



📐 Setting (copied from the paper, §1–§3)

(§1) Let A_j be the order generated by the j-th Galois orbit of columns, in the étale algebra K_j, and $\mathcal{O}_j$ its normalisation. The fusion ring sits in a chain $R \subseteq \prod_j A_j \subseteq \prod_j \mathcal{O}_j$. The last index is the gluing, $G(n,p) := v_p[\prod_j A_j : R]$. Writing $\mathcal{O}$ for an orbit of centred spectra under $\mathbb{F}_p^\times$ and $h_\mathcal{O}$ for its stabiliser order,

$$D = \sum_{\mathcal{O}} \frac{p-1}{h_\mathcal{O}}, \qquad b = \sum_{\mathcal{O}} 1, \qquad \Delta = \sum_{\mathcal{O}} \delta_\mathcal{O}, \qquad F_n(p) = \frac{1}{D} \sum_{\mathcal{O}} \frac{p-1}{h_\mathcal{O}} [\delta_\mathcal{O} \geq 2].$$

(§2) Part I reads the semigroup off the spectrum: with h the order of the multiplicative stabiliser of T and $e = (p-1)/h$ the ramification index, S_T is generated by e together with those $j<e$ for which $M_{hj}(T) \neq 0$. The defect $\delta(T) = \#(\mathbb{N}_0 \setminus S_T)$ is therefore a function of one thing only: *which power sums of T vanish*. *(§3)* For $J \subseteq \{1,2,3\}$ with $1 \in J$ write $C_J(n,p)$ for the number of centred T with $M_j(T) = 0$ for every $j \in J$. *(§5)* A subset of $\mathbb{F}_p^\times$ containing exactly one of each pair $\{a, -a\}$ is a *half-system*.

2 Where the defect comes from

Proposition 1. p. 5

> *At $n = 4$ and $h = 1$ the value $\delta = 2$ does not occur.*

↳ Not used by another result of the paper.

Proof

> First, $h = 1$ forces $e_3 \neq 0$: if $e_3 = 0$ then the centred quartic $X^4 + e_2X^2 - e_3X + e_4$ is even, so its root set is stable under $t \mapsto -t$ and $h \geq 2$. Hence $M_3 = 3e_3 \neq 0$ and $3 \in S_T$. Now if $M_2 \neq 0$ then $2 \in S_T$ too, so $S_T \supseteq \langle 2,3 \rangle$, whose only gap is 1, and $\delta = 1$. If $M_2 = 0$ then $e_2 = 0$, and by the cascade $M_5 = 0$ as well; hence $1, 2, 5 \notin S_T$ and $\delta \geq 3$. No spectrum is left for $\delta = 2$.

🛡️ Evidence & scope

- **Program:** Newton recurrence, the cascade $M_2 = 0 \Rightarrow M_5 = 0$ at $n = 4$, and zero orbits with $n = 4$, $\delta = 2$, over 7935 orbits, $p \le 31$, $3 \le n \le 7$. `cascada_newton.py`

Observation 2. p. 5

Let $h = 1$. Outside the four loci on which (M_2, M_3), (M_2, M_4), (M_2, M_5) or (M_3, M_5) vanish simultaneously, we find

$$\delta - 1 \;=\; [\,M_2 = 0\,] + [\,M_3 = 0\,].$$

This is measured, not proved, and everything below that rests on it inherits that status.

> NEW **Classical and new:** the paper labels it measured, not proved, and says nothing below depends on it.

↳ Not used by another result of the paper.

Evidence & scope

- **Program:** 438 orbits with $h = 1$ outside the loci, $p \le 19$, $3 \le n \le 7$, no exception. `tricotomia.py`

Corollary 3. p. 5

For $h = 1$, $\delta \ge 2 \iff M_2 M_3 = 0$.

↳ Used in: Theorem 5 (p. 5)

Proof

The corollary does not need the Observation; it follows from the generating rule alone. Since T is centred, $M_1 = 0$, so $1 \notin S_T$. If $M_2 M_3 \neq 0$ then $2, 3 \in S_T$, hence $S_T \supseteq \langle 2, 3 \rangle$, whose only gap is 1, and $\delta = 1$. If $M_2 = 0$ then $2 \notin S_T$: it is not a generator, and it cannot be a sum of smaller positive elements because the only candidate is 1, which is absent. So 2 is a second gap and $\delta \ge 2$. If $M_3 = 0$ the same argument applies to 3, whose only decomposition by smaller indices would use 1.

Lemma 4. p. 6

Let T be centred with $h = 1$ and $p > 7$. Then $\delta(T) = 3$ if and only if one of

$$\begin{array}{llll}
S_T = \langle 4,5,6,7 \rangle, & \text{gaps } \{1,2,3\}, & M_2 = M_3 = 0, & M_4 M_5 M_6 M_7 \neq 0; \\
S_T = \langle 3,5,7 \rangle, & \text{gaps } \{1,2,4\}, & M_2 = M_4 = 0, & M_3 M_5 M_7 \neq 0; \\
S_T = \langle 3,4 \rangle, & \text{gaps } \{1,2,5\}, & M_2 = M_5 = 0, & M_3 M_4 \neq 0; \\
S_T = \langle 2,7 \rangle, & \text{gaps } \{1,3,5\}, & M_3 = M_5 = 0, & M_2 M_7 \neq 0.
\end{array}$$

> NEW **Classical and new:** that there are exactly four semigroups of genus three is classical [RG09]; the paper claims the vanishing-moment half of each row.

↳ Used in: Theorem 8 (p. 5)

Proof

That there are exactly four numerical semigroups of genus three, and that they are the ones listed, is classical and not ours; the count by genus is catalogued [6] . For completeness: the multiplicity is at most four; if it is four the gaps are $1, 2, 3$; if three, the gaps include $1, 2$ and one more, which must be 4 or 5 because $\langle 3, 4, 5 \rangle$ contains every integer ≥ 3; if two, the gaps are the odd numbers below the smallest odd element, which must then be 7. What is ours is the second half of each row. The stated conditions are the generating rule read off each case: a minimal generator below e comes from a non-vanishing moment and a gap below e from a vanishing one, and the generators listed already produce every remaining index, so no further moment is constrained.

Evidence & scope

- **Program:** the four rows in both directions over every centred orbit with h = 1: p ≤ 19 all n, p = 23 with 4 ≤ n ≤ 9, p = 29 with 5 ≤ n ≤ 8; no fifth gap set. `familias_delta3.py`
- **Program:** the ladder of small defects (Figure 3). `escalera_defectos.py`

3 Almost every column has defect one

Theorem 5. p. 8

For every fixed $n \geq 5$,

$$\setlength\fboxsep6pt\setlength\fboxrule0.6pt\ F_n(p) \;=\; \frac{2}{p} - \frac{1}{p^2} \;+\; O_n\Big(p^{1-n/2}$$

and in particular $F_n(p) = 2/p + O_n\big(p^{-\min(2,\,n/2-1)}\big)$.

🆕 **Classical and new:** the sieve over partitions is that of Li and Wan [LW], as in Lai, Marino, Robinson and Wan [LMRW20]; only the estimate is used, never an exact count [GMPP23].

↳ Used in: Theorem 8 (p. 5), Theorem 9 (p. 6)

Evidence & scope

- **Program:** the four steps of the proof, each against the census (Step 3 exact for 5 ≤ n ≤ 8, p ≤ 43). `ley_2p.py`
- **Numerical illustration (not a proof):** the three-moment census at n = 8 up to p = 97: the coefficient −1 is not yet visible, as the paper says. `ley_2p_largo.py`, `ley_2p_estabilizadores.py`

Lemma 6. p. 8

Let $n<p$, let J be a set of positive integers with $1 \in J$ and $d = \max J<p$, and let $C_J(n,p)$ be the number of n-subsets of $\mathbb{F}_p$ with $M_j = 0$ for every $j \in J$. Then

$$\setlength\fboxsep6pt\setlength\fboxrule0.6pt\ \Big|C_J(n,p) - p^{-|J|}\tbinom{p}{n}\Big| \;\leq\; \big(1-p^{-|J|}\big)\,\frac{1}{n!}\prod_{r=0}^{n-1}\big((d-1)\sqrt{p}+r\big) \quad (1)$$

↳ Used in: Theorem 5 (p. 5), Theorem 8 (p. 5), Theorem 9 (p. 6)

Proof

Expand the $|J|$ conditions by additive characters. A frequency $(a_j)_{j\in J}$ contributes the sum of $e_p\big(\sum_{t\in T} f(t)\big)$ over n-subsets T, with phase polynomial $f(X) = \sum_{j\in J} a_j X^j$. The zero frequency gives $\binom{p}{n}/p^{|J|}$ exactly. On a non-zero frequency f is non-constant of degree at most d, so each character sum over $\mathbb{F}_p$ is at most $(d-1)\sqrt{p}$ by Weil. Expanding the n-subsets by cycle type contributes one such factor per cycle, and the identity $\sum_{\sigma\in S_n} x^{c(\sigma)} = x(x+1)\cdots(x+n-1)$ sums those contributions over S_n; dividing by $n!$ and summing the $p^{|J|}-1$ non-zero frequencies gives the stated bound.

Evidence & scope

- **Program:** the bound holds in all 48 cases of the three-moment census, n = 8, 29 ≤ p ≤ 97 (ratio bound/error 57 to $2.4\cdot10^5$). `cota_CJ.py`

Theorem 8. p. 11

For every fixed $n \geq 7$, with $\beta_n = \min(3, n/2-1)$,

$$\setlength\fboxsep{6pt}\setlength\fboxrule{0.6pt}\frac{\Delta(n,p)}{b(n,p)} = 1+\frac{2}{p}+\frac{3}{p^2}+O_n\big(p$$

and in the uniform measure on centred spectra $\mathbb{P}(\delta=3)=4/p^2+O_n(p^{-\beta_n})$.

↳ Used in: Theorem 9 (p. 6)

Proof

The inclusion–exclusion is over the four *loci* $L_{ab}=\{M_a=M_b=0\}$ with $ab\in\{23,24,25,35\}$, not over the four rows of Lemma 4: the rows carry non-vanishing conditions as well and describe four different semigroups, so as events they are *disjoint* and there is nothing to include or exclude. It is the loci that overlap, and by the consequence recorded after that lemma their union is exactly $\{h=1,\ \delta\ge 3\}$. Note that lying in L_{ab} is weaker than realising the corresponding semigroup: further moments may vanish and the defect may be larger.

Each locus imposes two vanishing conditions beyond $M_1=0$, so each has probability $p^{-2}+O_n(p^{1-n/2})$ by Lemma 6 with $d\le 5$. Any two of them together impose at least three — the six pairwise unions are $\{2,3,4\}$, $\{2,4,5\}$, $\{2,3,4,5\}$ and $\{2,3,5\}$ three times — so each intersection is $O_n(p^{-3}+p^{1-n/2})$, where the second term must be kept: at $n=7$ it is $p^{-5/2}$, which is larger than p^{-3}. Inclusion–exclusion gives $\mathbb{P}(\delta\ge 3)=4p^{-2}+O_n(p^{-\beta_n})$, and (7) with $r=4$ gives $\mathbb{P}(\delta\ge 4)=O_n(p^{-3}+p^{1-n/2})$. Since T is centred, $1\notin S_T$ whenever $h=1$, so $\delta=0$ forces $h>1$ and $\mathbb{P}(\delta=0)=O_n(p^{-3})$ for $n\ge 7$. Now

$$\mathbb{E}\,\delta=\big(1-\mathbb{P}(\delta=0)\big)+\mathbb{P}(\delta\ge 2)+\mathbb{P}(\delta\ge 3)+\sum_{r\ge 4}\mathbb{P}(\delta\ge r),$$

the last sum being at most $\binom{n-1}{2}\mathbb{P}(\delta\ge 4)$. Substituting Theorem 5 for the second term and the above for the third gives $1+2/p+(-1+4)/p^2$, and the change of measure adds nothing at this order.

Evidence & scope

- **Numerical illustration (not a proof):** at p ≤ 19, n ≤ 8 the ratio Δ/b is still between 0.50 and 1.01: the computed range neither confirms nor contradicts the theorem, which rests on its proof. `censo_delta.py`

Theorem 9. p. 12

Fix $0<\varepsilon<1/2$. *There is* $c_\varepsilon>0$ *such that, uniformly for* $\varepsilon p\le n\le(1-\varepsilon)p$,

$$\setlength\fboxsep{6pt}\setlength\fboxrule{0.6pt}F_n(p)=\frac{2}{p}-\frac{1}{p^2}+O_\varepsilon\big(e^{-c_\varepsilon p}\big)\ ,$$

Classical and new: the paper gives the proof as a sketch and says so.

↳ Not used by another result of the paper.

5 The centre: Gauss's half-systems

Proposition 10. p. 15

Let $T\subseteq\mathbb{F}_p^\times$ *satisfy* $M_r(T)=0$ *for every even* r *with* $2\le r\le p-3$. *Then* T *is empty, a half-system, or all of* $\mathbb{F}_p^\times$. *Conversely every half-system has those moments zero. In particular a non-empty proper* T *with those moments zero has* $|T|=(p-1)/2$: *the size is forced, not assumed.*

Classical and new: the paper has not found the statement anywhere, under five vocabularies, and attributes it to nobody.

↳ Used in: Proposition 11 (p. 7)

Proof

For a quadratic residue q put $w(q) = \#\{t \in T : t^2 = q\} \in \{0, 1, 2\}$. For $r = 2k$ the hypothesis reads $\sum_q w(q) q^k = 0$ for $1 \le k < (p-1)/2$. The full Vandermonde matrix on the distinct values q, with rows $k = 0, \dots, m-1$, is invertible, so the rows $k = 1, \dots, m-1$ have a kernel of dimension exactly one; and the constant vector lies in it, because $\sum_{q \in \mathrm{QR}} q^k = 0$ for $1 \le k < m$. Hence all $w(q)$ are equal, to 0, 1 or 2: T is empty, meets each pair $\{a, -a\}$ once, or is all of $\mathbb{F}_p^\times$. If $0 \in T$ the same argument applies to $T \setminus \{0\}$, which adds the half-systems with 0 adjoined, of size $(p+1)/2$.

Conversely, if T is a half-system then squaring is injective on T and T^2 runs over the quadratic residues exactly once, so for $r = 2k$ with $1 \le k < m$ one has $M_{2k}(T) = \sum_{s \in \mathrm{QR}} s^k = 0$.

Evidence & scope

- **Program:** no non-half-system with those moments over every orbit with p ≤ 19; the converse over 2924 half-systems. `semisistemas.py`

Proposition 11. p. 16

Let $p > 7$ be a prime with $(2 \mid p) = 1$ and let $m = (p-1)/2$. There is exactly one $k \in \{0, \dots, m\}$ with $8k(k+1) + 1 \equiv 0 \pmod p$, and for it the set

$$\setlength\fboxsep{6pt}\setlength\fboxrule{0.6pt} T_k = \{1, \dots, k\} \cup \{-(k+1), \dots,$$

is a centred half-system with trivial stabiliser and $S_{T_k} = \langle 3, 5, 7 \rangle$, hence $\delta(T_k) = 3$ and the order is not Gorenstein.

Classical and new: infinite explicit non-Gorenstein families are cheap (Marseglia [Mar22]); the paper claims only that this one is natural and its semigroup does not move with p.

↳ Not used by another result of the paper.

Proof

The congruence is $(2k+1)^2 \equiv \frac{1}{2}$, which has two roots in $\mathbb{F}_p$ because $(2 \mid p) = 1$; they are k and $p-1-k$, one on each side of m, and neither equals m, so exactly one lies in $\{0, \dots, m\}$. By construction T_k contains one of each pair $\{t, -t\}$, so it is a half-system and every even moment vanishes by Proposition 10. For the odd ones put $u = k(k+1)$ and $v = m(m+1)$, so that $u \equiv -\frac{1}{8}$ and $v \equiv -\frac{1}{4}$ in $\mathbb{F}_p$. Writing $F_j(t) = \sum_{r \le t} r^j$, one has $M_j(T_k) = 2F_j(k) - F_j(m)$ for odd j, and the classical closed forms $F_3 = \frac{[t(t+1)]^2}{4}$, $F_5 = \frac{[t(t+1)]^2(2t(t+1)-1)}{12}$ and $F_7 = \frac{[t(t+1)]^2(3[t(t+1)]^2 - 4t(t+1)+2)}{24}$ give, after substituting u and v,

$$M_3(T_k) = -\tfrac{1}{128}, \qquad M_5(T_k) = \tfrac{7}{1536}, \qquad M_7(T_k) = -\tfrac{245}{49152},$$

all non-zero for $p > 7$. The same two variables give the centring, which does not follow from the three moments above: $F_1(t) = t(t+1)/2$, so

$$M_1(T_k) \;=\; 2F_1(k) - F_1(m) \;=\; u - \tfrac{v}{2} \;\equiv\; -\tfrac{1}{8} + \tfrac{1}{8} \;=\; 0.$$

> So T_k is centred, and $M_3 M_5 M_7 \neq 0$ forces $h = 1$; the displayed semigroup then follows from the previous paragraph.

Evidence & scope

- **Program:** the 19 admissible primes p ≤ 200: half-system, $M_1 = 0$, h = 1, the three constants, gaps 1,2,4. `familia_Tk.py`

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